Deepfake Detection in Multimedia and Forensics: A Hybrid CNN Approach with EfficientNet and YOLO

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Abstract

Deepfake technology leverages deep learning to create highly realistic synthetic media, making it increasingly challenging to distinguish real content from manipulated images or videos. The rapid advancements in generative adversarial networks (GANs) have further enhanced the realism of deepfakes, raising serious concerns regarding misinformation, identity fraud, and digital forensics. Therefore, robust and efficient detection methods are crucial to mitigate the potential threats posed by deepfake content. In this paper, we propose a hybrid deep learning-based deepfake detection framework that integrates CNN-based EfficientNet and YOLO (You Only Look Once) algorithms to enhance the accuracy and speed of detection. The proposed approach follows a two-stage pipeline: (1) Face Detection using YOLO, which rapidly identifies and extracts facial regions from images or video frames, and (2) Deepfake Classification using EfficientNet, which applies a convolutional neural network (CNN) to extract facial features and classify whether the detected face is real or fake. The YOLO model employs a grid-based object detection strategy, predicting bounding boxes and confidence scores to accurately locate faces in images. Its loss function optimizes coordinate predictions and intersection-over-union (IoU) to ensure precise localization of facial regions. The extracted faces are then passed to EfficientNet, a CNN-based architecture optimized for computational efficiency and high accuracy. EfficientNet utilizes depth-wise convolutions, compound scaling, and Swish activation functions to improve feature extraction. The final classification layer applies a Softmax function to determine the probability of a face being real or deepfake.

To evaluate the performance of our proposed model, we utilize benchmark datasets such as FaceForensics++, Celeb-DF, and DeepFake Detection Challenge (DFDC). The model is trained using transfer learning techniques to enhance generalization across different deepfake manipulation techniques. Our results demonstrate that the EfficientNet-YOLO hybrid model outperforms conventional CNN-based classifiers in terms of accuracy, computational efficiency, and robustness against adversarial attacks.This study contributes to the field of deepfake detection by providing an end-to-end deep learning pipeline that effectively combines real-time face detection (YOLO) with high-accuracy classification (EfficientNet). The findings of this research hold significant implications for Cybersecurity, digital forensics, and social media content moderation, where the detection of manipulated media is crucial. Future work will focus on improving the model's resilience against adversarial attacks and expanding its applicability to video-based deepfake detection using temporal analysis techniques.

Keywords: Deepfake Detection, Deep Learning, CNN, EfficientNet, YOLO, Face Recognition, Digital Forensics, Cybersecurity.

Introduction

With the rapid advancement of artificial intelligence (AI) and deep learning, deepfake technology has emerged as a significant threat in digital media. Deepfake techniques leverage Generative Adversarial Networks (GANs) and Autoencoders to synthesize highly realistic fake images and videos, making it increasingly difficult for the human eye to distinguish between real and manipulated content. While deepfake technology has applications in entertainment and creative fields, it also raises serious concerns in cybersecurity, misinformation, identity fraud, and digital forensics. Traditional methods of deepfake detection relied on handcrafted features such as pixel inconsistencies, lighting variations, and facial asymmetry detection. However, these techniques often fail when dealing with high-quality deepfake videos that have been refined through adversarial training. To address these limitations, deep learning-based methods, particularly Convolutional Neural Networks (CNNs), EfficientNet, and YOLO, have been introduced to enhance the efficiency and accuracy of deepfake detection. In this study, we propose a hybrid deep learning-based deepfake detection model that combines YOLO (You Only Look Once) for real-time face detection and EfficientNet for high-accuracy deepfake classification.

The YOLO algorithm is utilized to detect and extract faces from images or video frames with high speed and precision. The extracted facial regions are then passed to EfficientNet, a CNN-based model that leverages compound scaling and optimized feature extraction to classify whether the face is real or fake [1]

Need for Deepfake Detection

The objective of this research is to develop a highly efficient and accurate deepfake detection system using CNN-based models, specifically EfficientNet and YOLO. The increasing threat posed by deepfake technology necessitates robust detection mechanisms that can identify manipulated images and videos with high precision. This study aims to enhance accuracy by leveraging EfficientNet for deepfake classification in images and YOLO for real-time video analysis, ensuring fast and reliable detection.

To improve the generalization of the model, a diverse dataset containing various types of deepfake manipulations will be used for training. Advanced *optimization techniques such as the Adam optimizer, RMSprop, and SGD will be applied to fine-tune the model and prevent overfitting. Additionally, data augmentation and binary cross-entropy loss functions will be incorporated to improve detection efficiency.

A key focus of this research is to achieve real-time deepfake detection by utilizing YOLO's fast object detection capabilities. The system will process videos with minimal latency, making it practical for real-world applications such as social media content verification, forensic analysis, [2] and cybersecurity. Furthermore, visualization techniques will be implemented to explain model predictions, providing insights into manipulated regions within an image or video frame.

EfficientNet and YOLO, this research aims to outperform traditional detection techniques, ensuring higher reliability, scalability, and real-world applicability in deepfake detection. The ultimate goal is to contribute to the fight against misinformation by providing an effective tool for identifying synthetic media.

Related Work and Findings

Liwei Deng, Hongfei Suo, Dongjie Li proposed a deepfake video detection system based on the EfficientNet-V2 network to combat the increasing misuse of deep learning in creating fake images and videos. Their system aims to enhance the accuracy of fake face detection by leveraging EfficientNet-V2's improved feature extraction capabilities. The proposed system was evaluated on two large-scale deepfake datasets, demonstrating superior performance compared to existing detection networks. The method effectively distinguishes real and fake faces while achieving high detection accuracy and strong visualization effects. However, the study has limitations, including potential challenges in detecting highly[3] sophisticated deepfakes generated using advanced generative models and the need for further improvements in real-time processing speed for large-scale applications.

Sohail Ahmed Khan, Duc-Tien Dang-Nguyen proposed a Hybrid Transformer Network for deepfake detection, leveraging an early feature fusion strategy to enhance detection accuracy. Their model integrates XceptionNet and EfficientNet-B4 as feature extractors, trained alongside a transformer network in an end-to-end manner using FaceForensics++ and DFDC benchmarks. The proposed system achieves comparable performance to state-of-the-art approaches while maintaining a relatively simple architecture. Additionally, the authors introduce novel face cut-out and random cut-out augmentations, which enhance the model's ability to generalize and reduce overfitting. However, limitations exist, including challenges in handling highly advanced deepfake manipulations and dependency on benchmark datasets, which may not [5] fully represent real-world deepfake scenarios.

Davide Alessandro Coccomini, Roberto Caldelli, Fabrizio Falchi, Claudio Gennaro, Giuseppe Amato conducted a cross-forgery analysis comparing Vision Transformers (ViTs) and EfficientNetV2 for deepfake image detection. Their study explores whether a deep learning technique can generalize well enough to detect manipulations from new deepfake generation methods without requiring frequent retraining. Using the ForgeryNet dataset, the proposed system evaluates EfficientNetV2 and Vision Transformers in detecting deepfake images. The findings indicate that EfficientNetV2 specializes in specific training methods, yielding high accuracy on known deepfake techniques, while Vision Transformers exhibit superior generalization, effectively detecting deepfakes generated using unseen methods. Limitations include challenges in dataset availability for training models on emerging deepfake techniques and potential computational costs associated with Vision Transformers.

Combining efficientnet and vision transformers for video deepfake detection Davide Alessandro Coccomini, Nicola Messina, Claudio Gennaro, Fabrizio FalchiInternational conference on image analysis and processing, Deepfakes are the result of digital manipulation to forge realistic yet fake imagery. With the astonishing advances in deep generative models, fake images or videos are nowadays obtained using variational autoencoders (VAEs) or Generative Adversarial [4] Networks (GANs). These technologies are becoming more accessible and accurate, resulting in fake videos that are very difficult to be detected. Traditionally, Convolutional Neural Networks (CNNs) have been used to perform video deepfake detection, with the best results obtained using methods based on EfficientNet B7. In this study, we focus on video deep fake detection on faces, given that most methods are becoming extremely accurate in the generation of realistic human faces. Specifically, we combine various types of Vision Transformers with a convolutional EfficientNet B0 used as a feature extractor, obtaining comparable results with some very recent methods that use Vision Transformers. Differently from the state-of-the-art approaches, we use neither distillation nor ensemble methods. Furthermore, we present a straightforward inference procedure based on a simple voting scheme for handling multiple faces in the same video shot. The best model achieved an AUC of 0.951 and an F1 score of 88.0%, very close to the state-of-the-art on the DeepFake Detection Challenge (DFDC). The code for reproducing our results is publicly available.[6]

Faysal Mahmud, Yusha Abdullah, Minhajul Islam, Tahsin Aziz proposed an explainable cost-sensitive deep learning approach for deepfake face detection in videos. Their system integrates four pre-trained CNN models—XceptionNet, InceptionResNetV2, EfficientNetV2S, and EfficientNetV2M—to ensure high accuracy and adaptability. Using FaceForensics++ and CelebDf-V2 as benchmark datasets, the model employs key frame extraction to process video data efficiently. A key innovation of this study is the application of a cost-sensitive neural network to address dataset imbalance, which frequently affects deepfake detection models. The XceptionNet model achieved 98% accuracy on CelebDf-V2, while InceptionResNetV2 obtained 94% accuracy on FaceForensics++, demonstrating the effectiveness of the proposed system. Limitations include [7] potential computational costs associated with deep learning models and challenges in detecting highly advanced deepfake manipulations.

Basma Yasser, Jumana Hani, Salma El-Gayar, Omar Amgad, Nourhan Ahmed, Hala M. Ebied, Habiba Amr, and Mohamed Salah proposed a deepfake detection system using EfficientNet-B4 and XceptionNet to differentiate real and fake videos. The study leverages the FaceForensics++ (FF++) and Celeb-DF (v2) datasets, employing frame extraction and face isolation techniques for preprocessing. The models are trained and evaluated using log loss and Area Under the Curve (AUC) metrics, demonstrating high accuracy in deepfake classification. The proposed system emphasizes the importance of continuously updating detection algorithms to counter evolving deepfake manipulation techniques. Limitations include potential computational complexity and performance variations on unseen deepfake methods, necessitating further improvements in generalization ability.

Mauro Barni, Patrizio Campisi, Edward J. Delp, Gwenaël Doërr, Jessica Fridrich, Nasir Memon, Fernando Pérez-González, Anderson Rocha, Luisa Verdoliva, Min Wu provide an extensive review of Information Forensics and Security (IFS), tracing its evolution over the past 25 years. The study highlights the role of IEEE Signal Processing Society (SPS) in advancing research and technological developments in data security, digital forensics, and intellectual property protection. It discusses key contributions in various domains of IFS, emphasizing the importance of ensuring data authenticity and holding perpetrators accountable. The paper also explores future trends in the field, addressing emerging challenges in digital security. Limitations include the complexity of adapting to rapidly evolving cyber threats and the increasing sophistication of adversarial attacks, necessitating continuous research and innovation

Falko Matern, Christian Riess, Marc Stamminger proposed a deepfake detection method that exploits visual artifacts left behind by facial editing algorithms. The study examines face tracking and editing artifacts commonly found in deepfake videos, particularly those generated by Deepfakes and Face2Face techniques. Their approach relies on simple yet effective visual features to expose manipulated faces, making the detection process interpretable even for non-technical users. The method demonstrates high adaptability, allowing for rapid adjustments to detect new manipulation techniques with minimal training data. Despite its simplicity, the approach achieves a strong AUC score of 0.866. Limitations include its reliance on visible artifacts, which might become less effective as deepfake generation methods improve, and difficulty in detecting high-quality [8] or adversarially modified deepfakes.

Lingzhi Li, Jianmin Bao, Ting Zhang, Hao Yang, Dong Chen, Fang Wen, Baining Guo introduced a novel Face X-ray technique for general face forgery detection. This approach generates a grayscale "X-ray" image that reveals whether an image is a blend of two different sources, distinguishing real faces from manipulated ones. The method capitalizes on the fact that most deepfake techniques involve a blending step, making Face X-ray a generalized detection approach that does not rely on specific deepfake generation artifacts. Extensive experiments demonstrate its robustness against unseen manipulation techniques, where traditional deepfake detection methods usually fail. Limitations include potential sensitivity to post-processing effects, such as noise and compression,[9] which may obscure blending boundaries, reducing detection accuracy in real-world applications.

Luca Guarnera, Oliver Giudice, Sebastiano Battiato proposed a Deepfake detection method by analyzing convolutional traces left by Generative Adversarial Networks (GANs) during the image generation process. Their technique uses an Expectation Maximization (EM) algorithm to extract a set of local features that capture the unique "fingerprint" of the underlying convolutional generative process. The method was tested on five different architectures (GDWCT, STARGAN, ATTGAN, STYLEGAN, STYLEGAN2) using the CELEBA dataset as ground truth for real images. Results showed high effectiveness in identifying different generative processes. However, limitations include sensitivity to post-processing techniques like compression and noise, which can obscure the convolutional traces, affecting detection accuracy.

Research Gap & Motivation

Despite significant advancements in deepfake detection, several critical gaps persist, necessitating further research and innovation. One major challenge is the limited generalization ability of detection models, as many struggle to accurately classify deepfakes generated by unseen methods, often overfitting to specific training datasets like FaceForensics++ and CelebDF-V2. Additionally, computational efficiency and real-time processing remain significant hurdles, with high-performing models [10] like EfficientNet-V2 and hybrid Transformer-CNN architectures often being too resource-intensive for large-scale deployment. Another pressing issue is the vulnerability of existing models to adversarial attacks and post-processing techniques, where methods relying on visual artifacts (e.g., Matern et al., 2019) or convolutional traces (e.g., Guarnera et al., 2020) are easily manipulated by compression, noise, or adversarial perturbations. Furthermore, the lack of standardized evaluation metrics and diverse real-world datasets impedes the ability to benchmark models effectively, as current datasets may not capture the evolving nature of deepfake manipulation. Additionally, interpretability and explainability remain underexplored, with many deepfake detection models operating as black boxes, making it difficult for forensic experts and regulatory bodies to validate their decisions. Given these challenges, this survey aims to provide a comprehensive review of existing deepfake detection methodologies, assessing their strengths and limitations while proposing strategies for improving generalization, computational efficiency, robustness, dataset diversity, and interpretability. By addressing these gaps, we aim to advance the development of more reliable, scalable, and explainable deepfake detection systems capable of effectively combating the growing threat of AI-generated forgeries in real-world applications.

Existing Deepfake Detection Techniques

Existing system have to detect Deepfake using inconsistent head poses. Algorithms used in the previous model create the face of different persons without changing the original expressions hence creating mismatched facial landmarks. The landmark locations of few false faces often vary from those of the real faces, as a consequence of interchanging faces in the central face region in the Deep Fake process. The difference in the distribution of the cosine distances of the two head orientation vectors for real and Deepfakes suggest that they can be differentiated based on this cue. It uses the DLib package for face detection and to extract 68 facial landmarks. The crucial point that can be inferred from this paper is the focus on how the

Deepfakes are generated by splicing a synthesized face region into the original image, and how it can also use 3D pose estimation for detecting synthesized videos.

Limitations of Existing Model

- Cosine Distances Limitation – It may fail to detect high-quality Deepfakes with minimal inconsistencies.
- DLib and 68 Facial Landmarks – It is sensitive to occlusions and lighting variations, reducing accuracy.
- 3D Pose Estimation – Advanced Deepfakes maintain head orientation, making detection unreliable
- It does not work for side faces and extreme non-frontal like looking down or up.

Proposed Deepfake Detection Model

The proposed system leverages CNN and EfficientNet for robust deepfake detection by extracting fine-grained facial features to differentiate real and fake images/videos. It processes frames from deepfake datasets like FaceForensics++, DFDC, and Celeb-DF, applying preprocessing techniques such as resizing, normalization, and data augmentation. EfficientNet enhances feature extraction with optimized depth, width, and resolution scaling, improving accuracy in detecting subtle artifacts like unnatural textures, boundary inconsistencies, and blending errors.

The trained CNN-EfficientNet model classifies images effectively, outperforming traditional methods by capturing intricate distortions in synthesized faces, ensuring better generalization across various deepfake types The training and evaluation of detection models are performed on these two large-scale fake face image The video-level dataset used is the face forensics datasets (FF++), which are widely used in the detection of various fake videos, while 140k Real and Fake Faces is the image-level dataset used. /video dataset.

Strengths of Proposed Approach

- EfficientNet follows the MBConv (Mobile Inverted Bottleneck Convolution) structure and scales three dimensions:
- Depth Scaling (More layers) →Extracts deeper features from deepfakes.
- Width Scaling (More channels per layer) →Captures more fine-grained features. Resolution Scaling (Higher input image size) → Retains more facial details for detection.
- Fast Processing – Convert videos to
- Image for quicker detection.
- Accurate Results – Scores faces from real (0) to fake (1).
- Strong Reliability – Trained on large, diverse datasets.

EXISTING VS PROPOSED SYSTEM

Classification	Existing System	Proposed System
Prediction	65.8	92
Module Processed	68	90.88
Image Processing	50	95
Video Processing	53.8	92.8

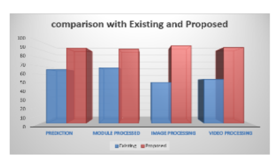

Fig 1 Existing and proposed system comparison

Research objectives based on deepfake detection

The primary objectives of this study are:

- To develop an efficient and robust deepfake detection framework that can process both images and video frames.

- To integrate YOLO for real-time face detection and EfficientNet for high-performance deepfake classification.
- To evaluate the model's performance using benchmark datasets such as FaceForensics++, Celeb-DF, and DFDC (DeepFake Detection Challenge).
- To analyze the mathematical foundations behind CNN-based deepfake detection and explore the differences between traditional methods and deep learning approaches.
- To provide insights into future improvements and real-world applications of deepfake detection in cybersecurity, law enforcement, and media authenticity verification.

Deepfake Detection Using YOLO and EfficientNet

The following research titles focus on deepfake detection using YOLO and EfficientNet as a hybrid framework. Hybrid Deepfake Detection using YOLO and EfficientNet presents a real-time approach for image and video forensics, while Deepfake Detection in Video Streams proposes a YOLO-based real-time framework with EfficientNet classification. Real-Time Deepfake Face Detection using YOLOv8 and EfficientNet emphasizes a deep learning-based approach for detecting manipulated media. Robust Deepfake Detection explores frequency domain analysis combined with YOLO-based face tracking. CNN-LSTM-Based Deepfake Detection integrates YOLO for face localization along with temporal analysis. Deepfake Video Detection leverages YOLOv7 and EfficientNet with optical flow motion estimation to enhance detection accuracy. Multi-Scale Deepfake Recognition proposes combining YOLO face detection and EfficientNet for improved results. Secure Digital Forensics introduces a deep learning framework utilizing YOLO and EfficientNet for digital media verification. Real-Time Deepfake Identification focuses on detecting fake media in social media platforms using YOLO for face detection and EfficientNet for classification. AI-Powered Deepfake Detection presents a YOLO and CNN-based hybrid model to ensure authenticity in video content.

Traditional Methods vs. CNN-Based (EfficientNet & YOLO) Deepfake Detection

Traditional deepfake detection methods rely on handcrafted features such as facial landmark inconsistencies, head movement analysis, and texture-based analysis. These approaches involve statistical and heuristic techniques to detect unnatural patterns in images and videos. However, they are often *limited by their inability to generalize across different types of deepfake manipulations, leading to lower accuracy and vulnerability to adversarial attacks.

On the other hand, CNN-based models like EfficientNet and YOLO leverage deep learning to automatically extract complex hierarchical features, making them more robust and adaptable to various deepfake styles. EfficientNet optimizes model efficiency by scaling depth, width, and resolution, making it effective for image-based deepfake classification. It employs binary cross-entropy loss and data augmentation to enhance performance and prevent overfitting. Meanwhile, YOLO (You Only Look Once) is well-suited for video-based deepfake detection, as it performs real-time object detection with bounding box prediction and evaluates results using Intersection over Union (IoU).

Unlike traditional methods, which require manual feature engineering, CNN-based models learn discriminative patterns automatically from large datasets, making them highly scalable. While traditional methods may still offer lightweight solutions for specific cases, CNN-based deepfake detection using EfficientNet and YOLO significantly improves detection accuracy, generalization ability, and real-time processing efficiency

Forensic and Security-Based Deepfake Research

Forensic and security-based deepfake research focuses on developing advanced methods to detect and prevent AI-generated media manipulation. Deepfake Forensics investigates various AI-powered video manipulation detection techniques to ensure the authenticity of digital content. Securing Digital Media proposes an AI-based deepfake detection system specifically designed for safeguarding online platforms against misinformation and tampered videos. YOLO-Based Deepfake Detection explores the application of real-time face detection and classification methods for cybersecurity purposes, particularly in social media and news platforms, to prevent the spread of fake content and ensure digital media integrity.

Hybrid Approaches and Ensemble Models

Combining multiple models and techniques has been a focus to enhance detection accuracy. Ensemble methods that integrate various deep learning models can capture diverse features indicative of deepfakes. For example, incorporating both spatial and temporal features through CNNs and Recurrent Neural Networks (RNNs) like Long Short-Term Memory (LSTM) networks has shown promise in detecting inconsistencies across video frames. A comprehensive review by Nguyen et al. (2019) discusses various deep learning-based approaches for deepfake detection, highlighting the importance of ensemble methods in improving detection robustness.

Architecture outline

The given architecture [fig 2] outlines a deepfake detection framework that integrates EfficientNet for image classification and YOLO for video-based deepfake identification. The process begins with uploading images or videos, which then undergo a pre-processing phase where videos are split into frames, faces are detected and cropped, pixel values are normalized, and the dataset is split into training (80%) and testing (20%) subsets. The processed data is then stored and loaded into two separate models: EfficientNet-B0 to B7 for image classification and YOLO for video-based analysis. During the training phase, EfficientNet utilizes the Adam optimizer, binary cross-entropy loss, and data augmentation techniques to improve accuracy while preventing overfitting. YOLO, on the other hand, focuses on bounding box detection and evaluates performance using Intersection over Union (IoU). Once trained, the model undergoes evaluation and testing, where precision, recall, and visualization techniques are used to assess its performance. The final model expectation outcome is to classify input images and videos as either real or deepfake with high accuracy and reliability. This structured approach ensures an efficient and effective deepfake detection pipeline by leveraging state-of-the-art deep learning techniques.

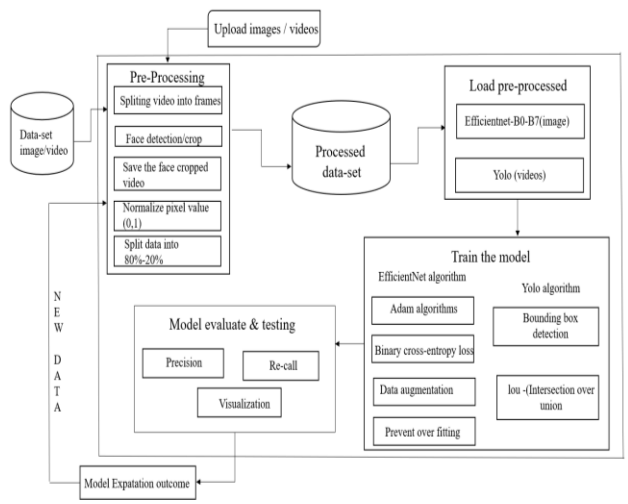

Figure 2 System Architecture

The provided architecture represents a deepfake detection framework using EfficientNet for image classification and YOLO for video-based deepfake identification. **Below is a step-by-step breakdown of the overall process:**

- **Upload Images/Videos.**

The system takes input in the form of images or videos that need to be analyzed for deepfake detection.

- **Pre-Processing Module.**

Splitting video into frames: Videos are broken down into individual frames for further analysis. Face detection and cropping: Faces are detected and extracted from frames using a detection algorithm .Save cropped faces: The detected face regions are stored separately. Normalize pixel values: The pixel values are scaled (e.g., to a range of 0 to 1) for better model efficiency. Split data into 80%-20%: The dataset is divided into training (80%) and testing (20%) sets.

- **Processed Data Storage**

After pre-processing, the dataset is stored for further analysis and training.

- **Load Pre-Processed Data**

EfficientNet-B0 to B7: Used for processing images for deepfake classification. YOLO (You Only Look Once): Used for video-based analysis, particularly face detection and tracking.

- **Training the Model**

EfficientNet Algorithm (for image-based deepfake detection) Adam optimizer: Used to optimize model learning. Binary cross-entropy loss: Applied for classification between real and fake images. Data augmentation: Enhances model performance by introducing variations in training data. Overfitting prevention: Regularization techniques are applied to ensure the model generalizes well.

YOLO Algorithm (for video-based detection)

The deepfake detection system employs bounding box detection to identify and localize faces in video frames, ensuring precise face extraction. IoU (Intersection over Union) is utilized to evaluate the accuracy of object detection by comparing the predicted bounding boxes with the ground truth. The model evaluation and testing phase uses precision and recall metrics to assess the effectiveness of the trained model, along with visualizations that provide insights into performance through graphical representations of results. The expected outcome of the system is to accurately classify images and videos as real or deepfake, ensuring high detection accuracy. This architecture establishes an end-to-end deepfake detection pipeline by integrating YOLO for video face localization and EfficientNet for image classification, enhancing the overall reliability and robustness of the system.

Process Architecture Flow

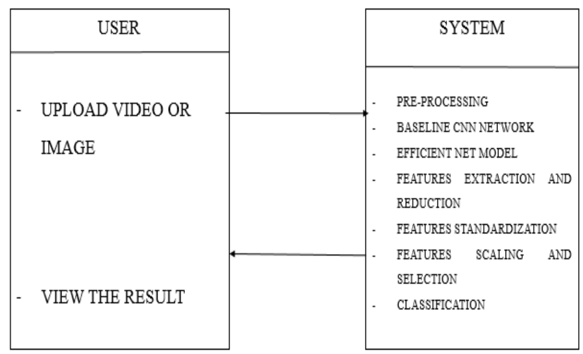

Figure (3)

The fig (3) presented in the UML class diagram illustrates a deepfake detection system with two primary components: User and System. The User interacts with the system by uploading a video or image, which is then processed through multiple stages within the System component. The System performs several operations, starting with pre-processing, followed by analysis using a baseline CNN network and EfficientNet model for feature extraction. The extracted features undergo reduction, standardization, scaling, and selection to optimize performance before passing through a classification module that determines whether the input is real or fake. Finally, the processed result is returned to the user. This structured approach ensures a robust and efficient deepfake detection framework by leveraging deep learning techniques to enhance accuracy and reliability.

Implementation Blocks

- Data preprocessing module
- Pre trained EEFICIENTNET model
- Train the model for deepfake classification
- Evaluate and test the model
- Result visualization and reporting

Data preprocessing module

The Data Preprocessing Module is essential for deepfake detection, ensuring high-quality input for deep learning models through face detection (using YOLO), resizing, normalization, and augmentation. It extracts frames from videos, crops and resizes detected faces (e.g., 224x224 for EfficientNet), and applies normalization for consistent feature extraction. Augmentation techniques like flipping, rotation, and brightness adjustments improve model robustness under various conditions. The module processes raw images or video files into preprocessed frames, optimizing them for further deepfake analysis.

Pre trained EEFICIENTNET model

The pre-trained EfficientNet model, combined with YOLO, enhances deepfake detection by efficiently classifying real vs. fake faces. YOLO detects and extracts facial regions from images or videos, which are then cropped, resized, and fed into EfficientNet for classification. This approach leverages YOLO's fast object detection and EfficientNet's advanced feature extraction to improve accuracy. By processing preprocessed image or video frames, the module outputs refined facial regions optimized for deepfake classification.

Train the model for deepfake classification

The deepfake classification model is trained by feeding preprocessed real and fake facial images into EfficientNet, leveraging large-scale datasets to optimize weights through backpropagation and gradient descent for improved accuracy. YOLO first extracts facial regions, which are then passed to EfficientNet to learn deepfake-specific features. After multiple training epochs, the model gains the capability to distinguish between real and deepfake images or videos with high precision. The final output is a probability score indicating whether the detected face is real or fake, aiding in decision-making processes.

Evaluate and test the model

The evaluation phase assesses the trained deepfake detection model on unseen real and fake face images/videos to measure its effectiveness using metrics like accuracy, precision, recall, F1-score, and AUC-ROC. The model is tested on diverse datasets to evaluate its generalization ability and robustness against various deepfake techniques, with final tuning performed to enhance detection accuracy and minimize false positives and false negatives. The classification process involves identifying True Positives (TP) when fake images are correctly classified, False Positives (FP) when real images are misclassified as fake, False Negatives (FN) when fake images are wrongly identified as real, and True Negatives (TN) when real images are correctly classified. To detect deepfake videos, the model aggregates per-frame predictions using EfficientNet and applies a majority vote or threshold-based approach (e.g., if 60% of frames are fake, the video is classified as deepfake), ultimately producing a single final decision for each video or image.

Result visualization and reporting

The Result Visualization & Reporting Module presents deepfake detection results through visual graphs, heatmaps, and classification scores, offering insights into model performance with metrics like accuracy, precision, recall, and confusion matrix. It enables users to interpret results using charts and graphs, enhancing the understanding of detection accuracy. Functionally, the module overlays bounding boxes and labels (Real/Fake) on detected faces in images/videos, displays the final classification result, and generates logs or reports for detected deepfakes, ensuring comprehensive performance analysis and clear reporting.

Data flow representation

The deepfake detection system follows a structured fig (4) workflow, beginning with data preprocessing, where face images/videos are extracted and normalized.

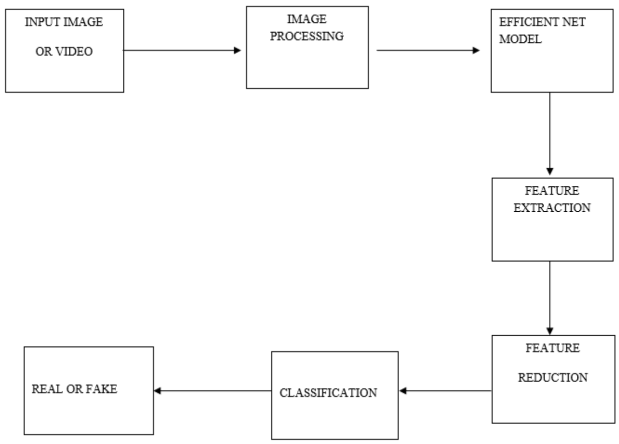

Figure (4)

The Evaluation & Testing Module then tests the trained model on unseen deepfake and real images/videos using metrics like accuracy, precision, recall, F1-score, and AUC-ROC. The model processes frame-level predictions using EfficientNet, aggregating results with majority voting or a threshold-based decision (e.g., if 60% of frames are fake, the video/image is classified as deepfake). After classification, the Result Visualization & Reporting Module overlays bounding boxes and labels (Real/Fake) on detected faces, generates performance reports, and visualizes key metrics like confusion matrix and heatmaps. The final results, including logs and detailed detection reports, are provided for user interpretation. Below is a flowchart representation:

Implementation of Detection Algorithms

Deepfake detection using CNN-based EfficientNet and YOLO (You Only Look Once) involves two key stages, Face Detection (YOLO) Identifies and localizes faces in images or video frames. Deepfake Classification (EfficientNet), Extracts features and classifies real vs. fake faces.

YOLO for Face Detection: Mathematical Explanation

YOLO is an object detection model that predicts bounding boxes and class probabilities for face detection. Bounding Box Prediction YOLO divides the input image into an S×S grid. Each cell predicts B bounding boxes with confidence scores.

Each bounding box consists of:

(x,y,w,h,c)

Where:

- (x,y)→ Center coordinates of the bounding box
- (w,h)→ Width and height of the bounding box
- c→ Confidence score (probability of containing a face)

The confidence score is:

c= P (face) × IoU(predicted box,ground truth)

Where:

P (face) is the probability of a face in the box.

IOU (Intersection over Union) measures the overlap between predicted and actual bounding boxes:

IOU = Area of Union / Area of Intersection

YOLO Loss Function

YOLO optimizes three main losses:

$$\mathcal{L} = \lambda_{\text{coord}} \sum_{i=0}^{S^2} 1_i^{\text{obj}} [(x_i - \hat{x}_i)^2 + (y_i - \hat{y}_i)^2]$$

$$+\lambda_{\text{coord}} \sum_{i=0}^{S^2} 1_i^{\text{obj}} [(w_i^{1/2} - \hat{w}_i^{1/2})^2 + (h_i^{1/2} - \hat{h}_i^{1/2})^2]$$

$$+\sum_{i=0}^{S^2} 1_i^{\text{obj}} (C_i - \hat{C}_i)^2 + \lambda_{\text{noobj}} \sum_{i=0}^{S^2} 1_i^{\text{noobj}} (C_i - \hat{C}_i)^2$$

where:

- 1_i^{obj} = 1 if object (face) exists, otherwise 0
- λ_{coord} and λ_{noobj} are hyperparameters to control loss weighting

Expected outcome

An input image or video frame containing a face is processed by YOLO, which divides the image into an S×S grid, where each grid cell predicts multiple bounding boxes and their confidence scores. Each bounding box is defined by its coordinates (x, y, w, h), representing the face's position and dimensions. Along with the bounding box, YOLO assigns a confidence score c, which indicates the probability of a face being present. To assess localization accuracy, the Intersection over Union (IoU) score is calculated, measuring the overlap between the predicted and ground-truth bounding boxes. The model optimizes its performance by minimizing three types of loss: Localization Loss, which ensures accurate bounding box placement, Confidence Loss, which refines face detection probability, and Classification Loss, which improves the distinction between objects within the detected bounding boxes.

EfficientNet for Deepfake Classification: Mathematical Explanation

EfficientNet is a Convolutional Neural Network (CNN) that classifies real vs. fake faces. Feature Extraction (Convolutional Layers) Each convolutional layer applies a filter W to the input X:

Where:

- X→ Input image tensor
- W→ Convolutional kernel (weights)
- b→ Bias
- ∗ → Convolution operation
- f(·) → Activation function (Swish activation)

Y=f (W∗ X+b)

Swish Activation:

$$f(x) = x \cdot \sigma(x) = \frac{x}{1 + e^{-x}}$$

Global Average Pooling (GAP) Layer

Instead of fully connected layers, EfficientNet uses GAP:

$$z_c = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} X_{i,j,c}$$

where $X_{i,j,c}$ is the feature activation for channel c.

Classification Layer (Softmax)

EfficientNet classifies the extracted features into real or fake using a Softmax function:

$$P(y = i | X) = \frac{e^{z_i}}{\sum_{j=1}^{N} e^{z_j}}$$

Where:

- z is the class score for class iii,
- N is the total number of classes,
- The output probability P(y) determines whether the face is **real or fake**.

Expected Result and Impact

Preprocessed face images detected by YOLO are fed into EfficientNet, where they pass through multiple convolutional layers to extract spatial and texture features essential for deepfake detection. These layers apply learned filters to capture patterns indicative of real or manipulated faces. The extracted features are then processed by fully connected layers, followed by a Softmax classifier that assigns probability scores for real or fake classification. If the probability surpasses a set threshold, the image is labeled as fake; otherwise, it is classified as real. Additionally, feature maps are generated to enhance interpretability, allowing visualization of critical regions that influenced the model's decision.

Feature extraction

In deepfake detection involves processing detected face images through convolutional layers to capture essential patterns distinguishing real and fake faces. EfficientNet applies convolutional filters to extract hierarchical features such as edges, textures, and deep structural attributes, refining them through activation functions like Swish for better gradient flow. The extracted feature maps undergo Global Average Pooling (GAP) to reduce dimensions while retaining critical information, followed by fully connected layers and a Softmax classifier, which assigns a probability score for real or fake classification. This process enhances the model's ability to detect subtle artifacts introduced by deepfake generation techniques, ensuring robust and accurate face verification.

Conclusion

Deepfake detection using YOLO for face localization and EfficientNet for classification is a powerful and scalable approach that enhances the accuracy and efficiency of detecting manipulated media. YOLO's grid-based detection enables rapid and precise facial extraction, while EfficientNet's deep convolutional layers analyze intricate features to differentiate between real and synthetic faces. The integration of **Swish activation, Softmax classification, and Global Average Pooling (GAP)** refines the feature extraction process, improving model generalization across diverse datasets. Despite these advancements, deepfake technology continues to evolve, necessitating ongoing research in adversarial training, explainable AI, and multimodal detection techniques to enhance robustness against sophisticated manipulations. Future improvements could focus on **real-time processing, adaptive learning models, and cross-modal verification methods** to strengthen deepfake detection in critical areas such as cybersecurity, social media regulation, and forensic investigations.

Future Enhancement

In the future, the proposed system can be enhanced by incorporating real-time deepfake detection capabilities to enable instant identification during live broadcasts or video streaming. The model can be further strengthened through adversarial training techniques, making it more robust against evolving and sophisticated deepfake generation methods. Additionally, integrating multimodal detection by analyzing audio, visual, and textual data together can improve accuracy and reduce false positives. Implementation of Explainable AI (XAI) methods can offer better transparency, allowing users to understand the decision-making process of the model. Expanding the dataset with more diverse and challenging deepfake samples will also contribute to better generalization and reliability of the system.

References

01. Deng, Liwei, Hongfei Suo, and Dongjie Li. "Deepfake Video Detection Based on EfficientNet-V2 Network." Computational Intelligence and Neuroscience 2022.1 (2022): 3441549.

02. Khan, Sohail Ahmed, and Duc-Tien Dang-Nguyen. "Hybrid transformer network for deepfake detection." Proceedings of the 19th international conference on content-based multimedia indexing. 2022.

03. Coccomini, Davide Alessandro, et al. "Cross-forgery analysis of vision transformers and cnns for deepfake image detection." Proceedings of the 1st International Workshop on Multimedia AI against Disinformation. 2022.

04. Coccomini, Davide Alessandro, et al. "Combining efficientnet and vision transformers for video deepfake detection." International conference on image analysis and processing. Cham: Springer International Publishing, 2022.

05. Mahmud, Faysal, et al. "Unmasking deepfake faces from videos using an explainable cost-sensitive deep learning approach." 2023 26th International Conference on Computer and Information Technology (ICCIT). IEEE, 2023.

06. Yasser, Basma, et al. "Deepfake Detection Using EfficientNet and XceptionNet." 2023 Eleventh International Conference on Intelligent Computing and Information Systems (ICICIS). IEEE, 2023

07. Verdoliva, Luisa. "Media forensics and deepfakes: an overview." IEEE journal of selected topics in signal processing 14.5 (2020): 910-932.

08Matern, Falko, Christian Riess, and Marc Stamminger. "Exploiting visual artifacts to expose deepfakes and face manipulations." 2019 IEEE Winter Applications of Computer Vision Workshops (WACVW). IEEE, 2019.

09. Li, Lingzhi, et al. "Face x-ray for more general face forgery detection." Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. 2020.

10. Guarnera, Luca, Oliver Giudice, and Sebastiano Battiato. "Deepfake detection by analyzing convolutional traces." Proceedings of the IEEE/CVF conference on computer vision and pattern recognition workshops. 2020.